

SEMINAR 3

History of AI

13th of April 2026

Andrzej (Andy) Szczepura

a.szczepura@vu.nl

Outline of this seminar

- Reading: Chapter 6 recap: The Maturing of the Mainframe
 - Reading: Chapter 7 recap: Real Time
 - Writing: ABC-structure
 - Writing: Working on assignment 3
- At the end of this session, you've recapped chapter 6 and 7 and completed assignment 3

Chapter 6 recap: The Maturing of the Mainframe

What is this chapter about?

What story does it tell, when does it begin and end, what are the main points in the narrative, what does it revolve around...?

Chapter 6 recap: The Maturing of the Mainframe

What is this chapter about?

How did the (US) computer business look like in the 60s? How did IBM become the leader of the industry?

Computer: how did adoption of the technology look like, what kind of computers were used in the 60s, how was the computer viewed by general public...?

Chapter 6 recap: The Maturing of the Mainframe

What is the narrative of the chapter?

Which events, people, institutions, machines... are discussed, what is their purpose in the narrative?

Chapter 6 recap: The Maturing of the Mainframe

What is the narrative of the chapter?

- *Cultural ambivalence (Desk Job, threat vs tool)*
- **IBM 1401**: fast printer attached (IBM 1403 – 4 times faster than popular IBM 407 accounting machine), quite cheap, Report Program Generator (RPG) → great sales (12000 vs 1000 forecasted)
- IBM and seven dwarfs – IBM has ~75% of the market, the rest shared by: Sperry Rand/UNIVAC, Honeywell, RCA, GE, Control Data, Burroughs, NCR. Different survival strategies:
 - IBM compatibility (RCA Spectra 70)
 - own compatible family (Honeywell, Burroughs, NCR)
 - niche specialisation (Control Data → scientific; GE → time-sharing).
- **System/360**: Software crisis (too many different, not compatible computers) causes IBM to design **New Product Line**. Big business risk, massive budget, and a huge success (but technically conservative).
- 1970–71 recession: RCA and GE exit; "IBM and the seven dwarfs" → IBM and the B.U.N.C.H.

Chapter 7 recap: Real Time

What is this chapter about?

What story does it tell, when does it begin and end, what are the main points in the narrative, what does it revolve around...?

Chapter 7 recap: Real Time

What is this chapter about?

Real time computing: it's origins and biggest projects which used it.

Chapter 7 recap: Real Time

What is the narrative of the chapter?

Which events, people, institutions, machines... are discussed, what is their purpose in the narrative?

Chapter 7 recap: Real Time

What is the narrative of the chapter?

- Batch processing vs real time computing – transformative technology, allowing much more use cases.
- **Project Whirlwind**: MIT aircraft trainer (1944) – Jay Forrester; first analog, then after meeting with Perry Crawford(1945), digital. Turns into a huge project, way above costs, almost cancelled. 1951 – Bill Papian presents core memory (replaces electrostatic memory by 1953). ONR founding ran out by 1951, but the project lives on as SAGE. (sidenote – route 128, east-coast tech hub, started largely due to Whirlwind).
- **SAGE**: Cold War air-defense network, 23 direction centers, IBM AN/FSQ-7 computers. Obsolete almost immediately (ICBMs), but a huge impact on the state of the technology (civilian spinoff projects, training lots of talent, drum memory, CRT culture...)

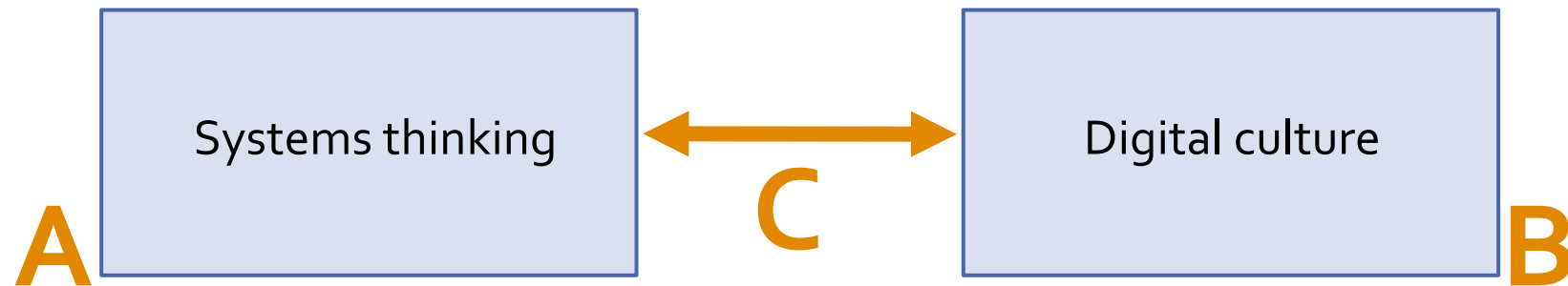
Chapter 7 recap: Real Time (contd.)

Civilian projects:

- **SABRE**: Airline reservation system (IBM + American Airlines, 1957-64). Greatly limits the logistical waste + transforms the entire airline business logic. 1,100 terminals in 50 cities. Architecturally similar to SAGE (Crawford involved here too!)
- **Credit cards & ATMs**: Diners Club (1949) → BankAmericard (1958) → Visa/NBI (1970) (led by Dee Hock).
BASE I (authorisation, 1973) and BASE II (clearing, 1974)
ATMs: Barclays London (1967), Chemical Bank NY (1969).
- **UPC**: Universal Product Code (1973). McKinsey (consulting) + IBM + a committee of vendors/manufacturers.
Political achievement as much as a technical one.

ABC-structure

To what extent does the concept of systems thinking fit the concept of digital culture?



- IF there is an A/B structure in the question:
 - 1) Explain A (with examples)
 - 2) Explain B (with examples)
 - 3) Explain C (the relation - make it explicit!)

Assignment 3.2. Real time

Campbell-Kelly attributes the development of real-time computer systems to a successful blending of military research projects and entrepreneurial efforts. Explain his line of thought.

- Refine the structure of your answer. Is it clear where the intro starts, where the body starts and where the conclusion starts?
- Did you answer all the questions?
- Exchange with neighbor for feedback.

Assignment 3.2. Real time

A

Campbell-Kelly attributes the development of real-time computer systems to a successful blending of military research projects and entrepreneurial efforts.

B Explain his line of thought.

- Refine the structure of your answer. Is it clear where the intro starts, where the body starts and where the conclusion starts?
- Did you answer all the questions?
- Exchange with neighbor for feedback.

C = B led to A (according to Campbell-Kelly)

Assignment 3.3. IBM and the seven dwarves

In the late 1950s and throughout the 1960s and 70s, IBM was dominating the computer manufacturing market. What role did other computer manufacturers play? Give an example of a relatively successful computer manufacturer and explain its' success.

- a) Refine the structure of your answer. Is it clear where the intro starts, where the body starts and where the conclusion starts?
- b) Did you answer all the questions?
- c) Exchange with neighbor for feedback.

QUESTIONS?
